HDR DATASET NOTE

Rotterdam tumour bank breast cancer cohort

HDR-NL-0001 | version 1.0.0 | 2026-11-11

DOI	Pending, minted at publication
CREATORS	Imad El Badisy
COVERAGE	Netherlands, Kingdom of the
PERIOD	1978 to 1993
UNIT	Patient
LICENCE	CC BY 4.0
LANDING PAGE	https://elbadisyimad.com/hdr/datasets/nl-rotterdam-breast-cancer/

Description

Records of 2,982 women with primary breast cancer operated on between 1978 and 1993 whose tumour tissue was stored in the Rotterdam tumour bank, the Netherlands: age, menopausal status, tumour size, grade, nodes, hormone receptors, adjuvant treatment, and time to relapse and to death.

A large cohort used to build breast cancer prognostic models. It is usually paired with the GBSG-2 trial for external validation.

GOOD FOR

- Developing and validating prognostic models
- Teaching Cox regression and non-linear effects
- Recurrence and death as competing or combined endpoints

NOT GOOD FOR

- Effects of modern treatments (most patients had no adjuvant therapy)

Origin

ORGANISATION	Erasmus MC Cancer Institute (Daniel den Hoed Cancer Center), Rotterdam
PUBLICATION	*Royston P, Altman DG (2013). External validation of a Cox prognostic model: principles and methods. BMC Medical Research Methodology 13: 33*
COUNTRY	Netherlands, Kingdom of the
ADDRESS	https://doi.org/10.1186/1471-2288-13-33
ACCESSED	2026-11-11
REUSE TERMS	Distributed openly for research in the survival R package (LGPL).
HOW OBTAINED	Taken from the BioBench R package (dataset rotterdam), which took them from survival::rotterdam.

Files

File	Content	Rows	Cols	Size
`rotterdam.csv`	One row per patient	2,982	15	149 kB
`nl-rotterdam-breast-cancer.xlsx`	All tables in one workbook, with About and Variables sheets			185 kB

SHA-256 checksums of every file are listed in `manifest.csv` on the landing page.

Variables

`rotterdam.csv` One row per patient

Variable	Type	Description
`pid`	integer	Patient identifier
`year`	integer	Year of surgery
`age`	integer	Age at surgery (years)
`meno`	integer	Menopausal status (0 premenopausal, 1 postmenopausal)
`size`	string	Tumour size in mm (<=20, 20-50, >50)
`grade`	integer	Differentiation grade (2 or 3)
`nodes`	integer	Number of positive lymph nodes
`pgr`	integer	Progesterone receptors (fmol/L)
`er`	integer	Oestrogen receptors (fmol/L)
`hormon`	integer	Hormonal treatment (1 yes, 0 no)
`chemo`	integer	Chemotherapy (1 yes, 0 no)
`rtime`	number	Days to relapse or last follow-up
`recur`	integer	Relapse (1 yes, 0 no)
`dtime`	number	Days to death or last follow-up
`death`	integer	Death (1 dead, 0 alive)

Known issues

Read these before analysis.

1. **Demonstration entry.** Taken from the BioBench R package to show how HDR presents a dataset. It has not gone through HDR review and will be removed before launch.
2. **Deaths after the end of recurrence follow-up.** 43 women died without recurrence after their recurrence censoring time, which affects how recurrence-free survival is built.
3. **Treatment was not randomised**, so hormon and chemo cannot be read as treatment effects.
4. **Grade has only two values** (2 and 3).

How to cite

Imad El Badisy (2026). Rotterdam tumour bank breast cancer cohort (version 1.0.0) [Data set]. HDR: Health Data for Research. https://elbadisyimad.com/hdr/datasets/nl-rotterdam-breast-cancer/

Please also cite the origin: Erasmus MC Cancer Institute (Daniel den Hoed Cancer Center), Rotterdam, Royston P, Altman DG (2013). External validation of a Cox prognostic model: principles and methods. BMC Medical Research Methodology 13: 33.

Versions

Version	Date	Changes
1.0.0	2026-11-11	Demonstration release

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